YI XIAO (肖懿)

Department of Computer Science and Technology,
Tsinghua University, P.R. China
email: yi.xiao.research@gmail.com | Google Scholar | GitHub

EDUCATION

Tsinghua University **Beijing, China**
Ph.D. Student, Computer Science and Technology (Expected Summer 2027) 2022 – Present
B.S., Mathematics and Physics 2018 – 2022

RESEARCH INTERESTS

Physical world modeling, generative models, dynamical systems, observation-guided state inference, data assimilation

SELECTED RESEARCH EXPERIENCE

Mohamed bin Zayed University of Artificial Intelligence **Abu Dhabi, UAE**
Project mentors/collaborators: Prof. Xiaosong Ma, Prof. Pascal Fua, Prof. Robert Jenssen Oct. 2025 – May. 2026

WindINR: Latent-State INR for Fast Local Wind Query and Correction in Complex Terrain

- Formulated local wind estimation as latent physical-state inference with a conditional INR, mapping terrain descriptors, kilometer-scale background forecasts, and continuous coordinates to high-resolution local wind estimates.
- Separated reusable representation learning from online state correction: fixed network weights and updated only a compact latent state using sparse station/UAV observations and a dataset-adaptive prior.
- Validated on Senja OSSEs, including a UAV-aided helicopter-approach scenario; improved RMSE over interpolation/fine-tuning baselines and achieved about 2.6x faster online correction than full-network fine-tuning.

Tsinghua University & Shanghai AI Lab (Joint Project) **Beijing & Shanghai, China**
Project mentors/collaborators: Prof. Wei Xue, Dr. Lei Bai, Dr. Ben Fei Nov. 2023 – Oct. 2025

FengWu-4DVar: AI-Embedded Variational Assimilation for Cyclic Weather Forecasting

- Coupled the differentiable FengWu AI weather model with 4DVar to build a self-contained cyclic forecast-assimilation system driven by observations.
- Used auto-differentiation to replace hand-coded adjoints and introduced spectral background-error approximation and temporal aggregation for efficient optimization.
- Demonstrated stable one-year cyclic forecasts under sparse simulated observations, with 6-hour assimilation in approximately 29 seconds on a single A100 GPU.

VAE-Var: Generative Priors for Variational Data Assimilation

- Designed a VAE-based non-Gaussian background-error prior and used the decoder as a nonlinear transform within variational data assimilation.
- Retained compatibility with arbitrary differentiable observation operators, enabling assimilation of sparse, irregular, and off-grid observations.
- Outperformed DiffDA, interpolation, and 3DVar on sparse ERA5 simulations; completed one-year cyclic assimilation with real GDAS prepbufr observations.

LoRA-EnVar: Parameter-Efficient Ensemble Variational Assimilation

- Proposed a hybrid generative EnVar framework that combines climatological VAE priors with flow-dependent ensemble perturbations via online LoRA adaptation.
- Updated only low-rank decoder adapters during each cycle, preserving the learned prior while avoiding full-decoder fine-tuning.
- Improved cyclic assimilation accuracy over static/generic hybrid baselines while reducing online trainable parameters by over three orders of magnitude.

Tsinghua University (HPCA Lab) **Beijing, China**
Advisor: Prof. Wei Xue Jan. 2021 – Nov. 2023

DNN-Integrated Ensemble Data Assimilation

- Built neural surrogate models for ensemble perturbation generation, reducing repeated ensemble-generation cost in ensemble-based data assimilation.
- Preserved analysis accuracy while substantially accelerating ensemble assimilation workflows.

PUBLICATIONS

FIRST-AUTHOR PUBLICATIONS & PREPRINTS

Xiao, Yi, Qilong Jia, Hang Fan, Pascal Fua, Robert Jenssen, Xiaosong Ma, Wei Xue.
WindINR: Latent-State INR for Fast Local Wind Query and Correction in Complex Terrain.
arXiv:2605.09511 (2026)

Xiao, Yi, Hang Fan, Kun Chen, Ye Cao, Ben Fei, Wei Xue, Lei Bai.
LoRA-EnVar: Parameter-Efficient Hybrid Ensemble Variational Assimilation for Weather Forecasting. In: **NeurIPS 2025**.

Xiao, Yi, Qilong Jia, Kun Chen, Lei Bai, Wei Xue.
VAE-Var: Variational Autoencoder-Enhanced Variational Methods for Data Assimilation in Meteorology. In: **ICLR 2025**.
(Earlier version on arXiv:2405.13711)

Xiao, Yi, Lei Bai, Wei Xue, Hao Chen, Kun Chen, Kang Chen, Tao Han, and Wanli Ouyang.
Towards a Self-Contained Data-Driven Global Weather Forecasting Framework. In: **ICML 2024**.
(Earlier version on arXiv:2312.12455)

SELECTED CO-AUTHORED PUBLICATIONS

Fan, Hang, Lei Bai, Ben Fei, **Yi Xiao**, Kun Chen, Yubao Liu, Yongquan Qu, Fenghua Ling, and Pierre Gentine.
Physically Consistent Global Atmospheric Data Assimilation with Machine Learning in Latent Space. **Science Advances**, 12(1): eaea4248, 2026.

Li, Yiyuan, Xiting Ju, **Yi Xiao**, Qilong Jia, Yongxiao Zhou, Simeng Qian, Rongfen Lin, et al.
Rapid Simulations of Atmospheric Data Assimilation of Hourly-Scale Phenomena with Modern Neural Networks. In: **SC 2023**.

CONFERENCE ABSTRACTS

Xiao, Yi, Qilong Jia, Kun Chen, Wei Xue, and Lei Bai.
Variational Autoencoder-Enhanced Variational Methods for Data Assimilation.
EGU25-14520, **Copernicus Meetings**, 2025.
(**Outstanding Student and PhD Candidate Presentation Award**)

Xiao, Yi, Lei Bai, Wei Xue, Kang Chen, Tao Han, and Wanli Ouyang.
Coupling the Data-Driven Weather Forecasting Model with 4D Variational Assimilation.
EGU24-2857, **Copernicus Meetings**, 2024

ACADEMIC SERVICES

Reviewer: ICML 2025, NeurIPS 2025, ICLR 2026, ICML 2026 (**Gold Reviewer**)

SELECTED AWARDS AND HONORS

2025	**EGU Outstanding Student and PhD Candidate Presentation Award**	May. 2025
2024-2025	**China National Scholarship**	Dec. 2024
2020-2021	**Academic Year Hesheng Award (Comprehensive Excellence)**	Dec. 2021
2019-2020	**Academic Year Hengda Award (Comprehensive Excellence)**	Dec. 2020
2018-2019	**Academic Year Academic Excellence Award**	Dec. 2019

ADDITIONAL INFORMATION

Technical Skills

- Programming: Python, C/C++, Java, MATLAB
- Parallel Computing: MPI, OpenMP, CUDA (GPU Programming)
- Tools: PyTorch, LaTeX
- Languages: Native in Mandarin, proficient in English